

SMoDO - Practical 10 - Stata

Dataset required: `SAheart.csv`

This and the following practicals make use of a dataset which are a subset of data from the Coronary Risk-Factor Study (CORIS) baseline survey. This was carried out in three rural areas of the Western Cape in South Africa. The participants were white males aged between 15 and 64, and the binary outcome of interest is whether they had myocardial infarction, as recorded in the `chd` variable. For our purposes, we will assume throughout that the `chd` outcome was ascertained during follow-up after the other variables were measured, but note that this is not how the data were actually obtained.

In this practical we will fit a logistic regression model for the CHD outcome and examine the performance of the fitted model using the techniques discussed in the lecture.

1. Load the dataset in Stata and fit a logistic regression model for the `chd` outcome with covariates `sbp tobacco ldl famhist obesity alcohol age`. What do the model results suggest about the associations between these covariates and the `chd` outcome?
2. Generate the predicted probabilities of CHD from the fitted model, and create a new variable which forms a 0/1 prediction of the outcome based on whether the predicted probability exceeds 0.5 or not. Based on this, estimate the accuracy of the fitted model.
3. Use `lsens` to examine the sensitivity vs. specificity for different cut-offs of the predicted probability. Suppose the covariates were to be used to identify men who had a high risk of a future CHD event in order to give further consultation with advice on lifestyle modifications or possible treatments to reduce future CHD risk. In this context, based on the plot from `lsens` what risk cut-off would you recommend?
4. Use `lroc` to estimate the area under the ROC curve. What do you conclude about the predictive ability of the model?
5. Use `estat gof` to perform Hosmer-Lemeshow's test with 10 groups. What conclusion do you draw?
6. Generate the fitted probabilities from the model and use these to assess model calibration visually using `lowess`. What do you conclude?
7. Call the following code to split the sample into training and test subsets:

```
splitsample, gen(svar, replace) split(0.75 0.25) show rseed(723442)
```

Fit the logistic regression using the training subset and estimate the AUC using this subset. Then estimate the AUC of the same fitted model using the test subset. What do you find, and is it in line with expectations?

8. Use the following code to run 10-fold cross-validation to estimate the AUC:

```
splitsample, gen(svar, replace) nsplit(10) rseed(723442)

postfile auccv svar auc using auccv, replace

forvalues i=1(1)10 {
    *fit model using all observations except i'th fold
    logit chd sbp tobacco ldl i.famhist obesity alcohol age if svar!='i'
    *estimate AUC on i'th fold
    lroc if svar=='i', nograph
    *post AUC to file
    post auccv ('i') (r(area))
}

postclose auccv
```

Having run the code, load the 'auccv' dataset it should have created and examine the AUC estimates across the folds. What conclusions do you draw from the results?